DETERMINANTS OF STUDENT ACADEMIC PERFORMANCE:

A DATA ANALYSIS APPROACH

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INTRODUCTION

Student academic performance is influenced by a combination of behavioral, academic, and socio-economic factors. Understanding these determinants is essential for improving learning outcomes and designing effective educational strategies.

This project analyzes the key drivers of student performance using a dataset of 6,378 observations. The objective is to examine how factors such as study habits, attendance, motivation, and prior academic achievement influence exam outcomes.

The analysis applies the Ordinary Least Squares (OLS) regression model to quantify relationships between variables, alongside data visualization techniques to identify patterns and trends. Python is used for data cleaning and econometric analysis, while Power BI is used to develop an interactive dashboard for visual insights.

The findings provide data-driven insights that can support students, educators, and policymakers, while also demonstrating practical skills in data analysis, statistical modeling, and data visualization.

DATA DESCRIPTION

The dataset used in this study consists of 6,378 observations and includes variables that capture academic performance, student behavior, and background characteristics. The dependent variable is **Exam_Score**, which measures students' final academic performance.

The independent variables are grouped into 3 categories:

1. **Academic-related Factors**
    - Hours Studied
    - Attendance
    - Previous Exam Scores
2. **Behavioral Factors**
    - Motivation Level

3. **Environmental Factors**
    - Parental Involvement
    - Family Income
    - Teacher Quality
    - Access to Resources

In addition, binary variables such as Gender and School Type are included to account for demographic differences. The final dataset used for analysis contained no missing values and all variables were encoded into appropriate numerical formats for statistical modeling.

METHODOLOGY

This study employs the Ordinary Least Squares (OLS) regression model to examine the relationship between student academic performance and its determining factors. The OLS method is used to estimate the effect of each independent variable on the dependent variable while holding other factors constant. This method ensures unbiased and efficient parameter estimates.

The general form of the regression model is specified as follows:

Exam _Score = β_0 + β_1Hours_Studied + β_2 Attendance + β_3 Motivation _Level + β_4 Previous _Scores +…+ ε

Where β_0 represents the intercept, β_1, β_2 , … are the coefficients measuring the impact of each explanatory variable, and ε is the error term.

After estimating the model, diagnostic tests were conducted to assess the validity of the regression assumptions, including tests for multi collinearity, heteroscedasticity and autocorrelation.

RESULTS AND INTERPRETATION

The regression results indicate that the model explains a significant portion of the variation in student performance with an R^2 value of 0.721. This suggests that approximately 72.1% of the variation in exam scores is explained by the variables included in the model.

The overall model is statistically significant, indicated by the F-statistic (p-value 0.000), implying that the explanatory variables jointly have a significant effect on student performance.

Key variables such as Hours _Studied, Attendance, Motivation_ Level and Previous_ Scores were found to have a positive and statistically significant impact on exam scores. This indicates that increased study time, higher attendance, stronger motivation and better prior academic performance are associated with improved outcomes.

Sleep_ hours, Gender and School_ Type do not have a significant effect on exam score since the p values are less than 0.05. They do not have a meaningful impact on student performance within this dataset.

For example, the coefficient for Hours _Studied (0.2948) suggests that a one – unit increase in study hours leads to an approximate increase of 0.29 marks in exam score, holding other factors constant.

Similarly, Attendance shows a strong positive effect, indicating that consistent participation contributes significantly to academic success.

The magnitude and statistical significance of these variables highlight the importance of both effort-related and behavioral factors in determining academic performance.

DIAGNOSTIC TESTS

To ensure validity of the regression results, several diagnostic tests were conducted to verify the assumptions of the OLS model.

Variance Inflation Factor (VIF) was used to test for multi collinearity among the independent variables. From the results, all VIF values were close to 1, indicating the absence of multi collinearity.

Breusch -Pagan test was applied to detect heteroscedasticity. The test yielded a p-value greater than 0.05 which indicates no evidence of heteroscedasticity and that the variance of the error term is constant

Durbin- Watson statistic was approximately equal to 2, indicating no presence of autocorrelation in the residuals.

These results confirm that the OLS assumptions are satisfied and the estimated model is reliable for inference.

DASHBOARD INSIGHTS

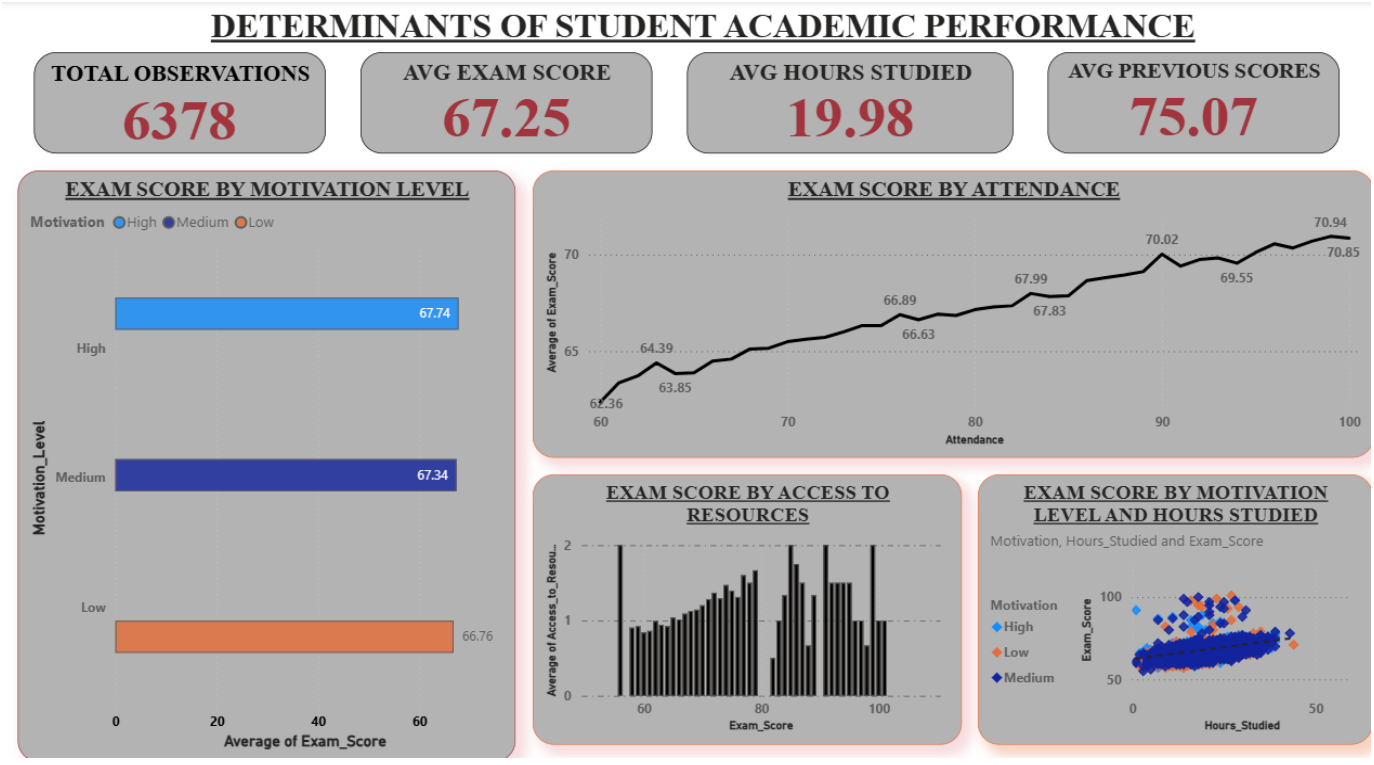


Figure 1: Student Performance Dashboard.

The Power BI dashboard provides a visual representation of the key factors influencing student academic performance.

The analysis shows that study hours and motivation level have a strong combined effect on exam performance, with highly motivated students achieving better outcomes even at similar levels of study time. This emphasizes the importance of both effort and mindset in academic success.

Additionally, attendance exhibits a positive relationship with exam scores. This indicates that students who attend classes more consistently tend to perform better.

The dashboard also highlights the consistency in academic ability over time since students with higher previous scores tend to maintain strong academic performance

The use of interactive visualization enhances the interpretability of the results and allows for deeper exploration of the relationships between variables.

CONCLUSION AND RECOMMENDATIONS

The results of this study indicate that student performance is primarily influenced by effort-related and behavioral factors, particularly study hours, attendance, motivation level and previous academic achievement.

The findings also show that increased study time, higher levels of motivation and consistent attendance are associated with improved exam outcomes. Additionally, prior academic performance plays a significant role, suggesting that learning outcomes are often consistent over time.

Based on these findings, several recommendations can be made:

I. Students should focus on improving study habits such as goal-setting and time management strategies. Maintaining consistent attendance and developing strong motivation also enhance academic performance.
II. Educational institutions should implement strategies that encourage student engagement such as mentorship programs, tutoring and continuous assessment.
III. Parents and guardians should increase their involvement in students' academic activities, this can positively influence academic outcomes.
IV. Policy makers should consider investing in resources and support structures that create a conducive learning environment for students such as improving access to learning materials and quality teaching.

TOOLS USED

- Python (Pandas, Statsmodels) for data cleaning and regression analysis.
- Power BI for data visualization and dashboard development.